

Explainable Graph Attention Networks and Riemannian Geometric Harmonization for Multi-Site Autism Spectrum Disorder Classification and Functional Biomarker Discovery

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Abstract—Autism Spectrum Disorder (ASD) is a complex neurodevelopmental condition characterized by socio-communicative atypicalities and restricted behavioral patterns. While resting-state functional Magnetic Resonance Imaging (rs-fMRI) enables non-invasive connectomic investigation, automated multi-site clinical diagnosis is severely constrained by cross-scanner acquisition heterogeneity and black-box algorithmic opacity. In this work, we present an explainable geometric deep learning and harmonization framework evaluated on the multi-site ABIDE I dataset (N=871, 17 international scanning centers). First, we design an Improved Graph Attention Network (Improved GAT) combining multi-head topological self-attention with dual global graph pooling (Mean + Max) to capture individualized connectome representations directly from 116 Automated Anatomical Labeling (AAL) regions. Second, to provide clinician-verifiable insights, we deploy GNNExplainer, isolating top-15 diagnostic functional subgraphs and quantitatively corroborating the Long-Range Inter-Lobar Dysconnectivity Hypothesis (85% of salient predictive edges connect distant anatomical lobes). Third, to eliminate cross-scanner hardware confounding, we introduce a Riemannian Tangent Space Geometric Harmonization pipeline with ComBat empirical Bayes batch correction. Multi-paradigm benchmarking demonstrates that while standard SVM collapses to 1.64% sensitivity (54.20% accuracy), Improved GAT achieves balanced clinical sensitivity (57.79%), and ComBat Tangent SOTA establishes a state-of-the-art benchmark of 68.88% Accuracy (0.755 AUC-ROC) across all 871 subjects, advancing to 69.72% Accuracy (0.766 AUC-ROC) on major sites (N=450+) and 70.35% Accuracy on single-site standardized cohorts (N=184). This work delivers a leak-free, interpretable, and clinically actionable framework for neurodevelopmental precision medicine.

Keywords—Autism Spectrum Disorder, Resting-State fMRI, Graph Attention Networks, Explainable Artificial Intelligence, Riemannian Manifold Harmonization.

I. INTRODUCTION

Autism Spectrum Disorder (ASD) represents a pervasive, highly heterogeneous neurodevelopmental condition characterized by persistent atypicalities in socio-emotional reciprocity, non-verbal communicative gestures, atypical sensory reactivity, and restricted, repetitive behavioral repertoires [1]. According to recent global epidemiological statistics, ASD affects approximately 1 in 36 children worldwide, imposing profound, lifelong challenges upon individuals, families, and healthcare infrastructures. Early objective screening and individualized therapeutic interventions have been unequivocally demonstrated to optimize long-term cognitive and socio-communicative trajectories. Nevertheless, contemporary gold-standard psychiatric evaluations rely predominantly upon observational schedules (such as the Autism Diagnostic Observation Schedule, ADOS-2 [17]) and

retrospective developmental interviews (e.g., ADI-R). While clinically essential, these behavioral assessments exhibit notable practical drawbacks: they are labor-intensive, require extensive specialized clinical training, exhibit examiner subjectivity, and cannot detect subtle micro-connectomic vulnerabilities before overt behavioral signs emerge.

To establish objective, biologically grounded diagnostics, resting-state functional Magnetic Resonance Imaging (rs-fMRI) has emerged as a premier non-invasive imaging modality. By tracking spontaneous, low-frequency fluctuations in the Blood-Oxygen-Level-Dependent (BOLD) signal across the resting brain, functional connectivity (FC) measures the temporal synchrony between spatially distributed Regions of Interest (ROIs) [1], [13]. Extensive neuroimaging investigations have demonstrated that individuals with ASD exhibit widespread functional disruptions spanning the Default Mode Network (DMN), Salience Network (SN), Frontoparietal Control Network (FPN), and primary sensorimotor circuits [19]–[21].

The Autism Brain Imaging Data Exchange I (ABIDE I) consortium aggregated 871 quality-controlled rs-fMRI acquisitions across 17 independent international neuroimaging centers [1]. However, developing machine learning classifiers on multi-site connectomics presents three fundamental challenges:

- **Multi-Site Acquisition Heterogeneity:** Scans in ABIDE I were acquired using disparate MRI scanner manufacturers (Siemens, Philips, GE), differing field strengths (1.5T vs 3.0T), diverse head coils, variable repetition times (TR ranging from 1.5s to 3.0s), and heterogeneous participant instructions (eyes open vs eyes closed). These discrepancies introduce pronounced inter-site batch effects that often dominate true neurobiological variance [8], [9].
- **Non-Euclidean Connectome Geometry:** Functional connectivity covariance matrices are Symmetric Positive Definite (SPD) structures lying on a curved Riemannian manifold. Conventional machine learning algorithms that flatten covariance matrices into Euclidean feature vectors violate this underlying Riemannian geometry, causing severe geometric distortion and poor generalization [5], [6].
- **The Interpretability Crisis:** Deep neural networks operate as opaque 'black boxes.' In clinical medicine, high accuracy alone is insufficient; practitioners require mechanistic interpretability identifying precisely which anatomical circuits drove each diagnostic decision [4], [11].

To resolve these challenges, this study presents a holistic multi-paradigm framework: (1) an Improved Graph Attention

Network (Improved GAT) capturing individualized topological brain graphs; (2) post-hoc Explainable AI (XAI) via GNNExplainer validating the Long-Range Dysconnectivity Hypothesis [14], [15]; and (3) a Riemannian Tangent Space Harmonization pipeline with ComBat batch correction achieving state-of-the-art diagnostic accuracy across multi-site and single-site cohorts [6], [8].

II. II. RELATED WORK

A. A. Traditional Machine Learning and Deep Learning Baselines

Early computational approaches to ASD classification on ABIDE relied upon vectorizing Pearson correlation matrices into high-dimensional feature vectors ($116 \times 115/2 = 6,670$ edges) evaluated via Support Vector Machines (SVM with linear and RBF kernels) or Random Forests [12], [13]. While these methods demonstrated reasonable performance within isolated single-site cohorts (such as the New York University Langone Medical Center cohort, $N = 184$), they uniformly suffered dramatic performance drops when evaluated across the full multi-site ABIDE I dataset. Crucially, as we demonstrate empirically in this work, standard SVM models trained on multi-site data succumb to majority-class bias: due to subtle class imbalance and severe cross-site noise, the decision boundary collapses to predicting the majority neurotypical control class, resulting in near-zero clinical sensitivity (1.64%).

B. B. Deep Learning and Convolutional Neural Networks

To extract non-linear representations, researchers explored deep learning architectures, including deep autoencoders (Heinsfeld et al. [12]) and hybrid convolutional frameworks such as ASD-DiagNet (Eslami et al. [16]). Although deep learning models demonstrated enhanced representation capacity, standard CNNs enforce translation invariance across adjacent grid pixels. This spatial inductive bias is intrinsically flawed for brain connectomics, where ROI ordering in an adjacency matrix is arbitrary and spatial proximity on a 2D matrix does not correspond to anatomical or topological proximity in the brain.

C. C. Graph Neural Networks and Population Graphs

Graph Neural Networks (GNNs) provide a natural, mathematically principled framework for modeling brain networks by representing anatomical regions as nodes $\mathcal{V}$ and functional connections as edges $\mathcal{E}$. Kipf & Welling [3] introduced Graph Convolutional Networks (GCN) using localized first-order approximations of spectral graph convolutions. Pariset et al. [7] pioneered the Population GCN (ParIS-GCN) framework, constructing a single meta-graph where nodes represent entire subjects and edges encode phenotypic and demographic similarities (age, sex, acquisition site). However, population graphs remain vulnerable to inter-scanner discrepancies because message passing across cross-site edges introduces substantial confounding noise.

D. D. Riemannian Geometry and Scanner Harmonization

Because covariance matrices are Symmetric Positive Definite (SPD), Riemannian geometry provides an optimal mathematical framework for functional connectomics [5], [6]. Varoquaux et al. [5] and Dadi et al. [6] proved that projecting covariance matrices onto the Euclidean tangent space anchored at the Riemannian Fréchet mean linearizes manifold curvature, preserving true geodesic distances. Furthermore, empirical

Bayes ComBat harmonization [8], [9] removes inter-scanner location and scale shifts while strictly preserving biological covariates. In this study, we synthesize Riemannian tangent space geometry with ComBat harmonization, demonstrating that geometric linearization combined with batch correction establishes a superior high-accuracy benchmark.

III. III. MATERIALS AND METHODOLOGY

A. A. Dataset Preprocessing and Brain Graph Construction

We evaluated $N = 871$ subjects (403 ASD, 468 Typical Controls) from ABIDE I preprocessed via the Configurable Pipeline for the Analysis of Connectomes (CPAC). Preprocessing steps included slice-timing correction, 3D motion realignment, 24-parameter Friston nuisance regression, white matter and CSF signal regression, temporal bandpass filtering (0.01 – 0.10 Hz), and non-linear spatial normalization into MNI152 stereotactic space. Regional BOLD time series were extracted using the Automated Anatomical Labeling (AAL-116) atlas [10], which parcellates the brain into $P = 116$ distinct anatomical ROIs.

TABLE I. TABLE I. ABIDE I MULTI-SITE COHORT CHARACTERISTICS AND ACQUISITION PROTOCOLS

Site ID	Scanner Vendor	Field	TR (s)	TE (ms)	N Total	ASD / Ctrl	Age (Mean $\pm$ SD)
NYU	Siemens Allegra	3.0T	2.00	15	184	79 / 105	15.7 $\pm$ 6.2
UCLA_1/2	Siemens Trio	3.0T	3.00	28	110	54 / 56	13.1 $\pm$ 2.6
PITT	Siemens Allegra	3.0T	1.50	25	57	29 / 28	18.9 $\pm$ 6.8
USM	Siemens Trio	3.0T	2.00	28	71	43 / 28	21.6 $\pm$ 8.4
TRINITY	Philips Achieva	3.0T	2.00	28	49	24 / 25	17.4 $\pm$ 3.8
YALE	Siemens Trio	3.0T	2.00	25	56	28 / 28	12.8 $\pm$ 2.9
LEUVEN	Philips Intera	3.0T	1.67	33	64	29 / 35	18.1 $\pm$ 4.7
CALTECH	Siemens Trio	3.0T	2.00	30	38	19 / 19	27.3 $\pm$ 10.4
OTHER (9 Sites)	Various Vendors	1.5/3.0T	1.5-3.0	25-35	242	98 / 144	16.4 $\pm$ 7.1
GRAND TOTAL	17 Scanning Centers	1.5 - 3.0T	1.5 - 3.0	15 - 35	871	403 / 468	17.0 $\pm$ 7.6

B. B. Node Feature Engineering and Graph Topology

For each subject $i \in \{1, \dots, N\}$, pairwise functional connectivity was computed via Pearson product-moment correlation between BOLD time series. An individualized graph $\mathcal{G}_i = (\mathcal{V}, \mathcal{E}_i, \mathbf{X}_i^{\text{node}}, \mathbf{E}_i^{\text{attr}})$ was constructed with: (1) Nodes $\mathcal{V}$: 116 AAL anatomical ROIs; (2) Node Features $\mathbf{X}_i^{\text{node}} \in \mathbb{R}^{116 \times 6}$: six higher-order statistical moments capturing BOLD temporal dynamics (temporal mean μ , standard deviation σ , skewness γ_1 , kurtosis γ_2 , total spectral power $\mathcal{P}$, and temporal variance σ^2); (3) Edges $\mathcal{E}_i$: top- k absolute correlation sparsification ($k = 10$ nearest neighbors per ROI); and (4) Edge Attributes $\mathbf{E}_i^{\text{attr}} \in \mathbb{R}^{|\mathcal{E}_i| \times 1}$: continuous signed Pearson correlation coefficients $r_{uv} \in [-1, 1]$.

C. C. Improved Graph Attention Network (Improved GAT) Architecture

The Graph Attention Network applies multi-head self-attention to dynamically weigh neighbor ROI communications. For layer l and attention head $k \in \{1, \dots, K\}$, the normalized attention coefficient α_{uv}^k between node u and neighbor $v \in \mathcal{N}(u)$ is defined as:

α_{uv}^k

$$= \frac{\exp\left(\text{LeakyReLU}(\mathbf{a}_k^T [\mathbf{W}_k \mathbf{h}_u^{(l)} \parallel \mathbf{W}_k \mathbf{h}_v^{(l)} \parallel \mathbf{W}_e \mathbf{e}_{uv}])\right)}{\sum_{w \in \mathcal{N}(u)} \exp\left(\text{LeakyReLU}(\mathbf{a}_k^T [\mathbf{W}_k \mathbf{h}_u^{(l)} \parallel \mathbf{W}_k \mathbf{h}_w^{(l)} \parallel \mathbf{W}_e \mathbf{e}_{uw}])\right)}$$

where $\mathbf{W}_k \in R^{D_{\text{out}} \times D_{\text{in}}}$ is a parameterized weight matrix, $\mathbf{a}_k \in R^{2D_{\text{out}} + D_e}$ is the attention vector, and $\parallel$ denotes concatenation. The node representation is updated by aggregating multi-head attended neighbor representations:

$$\mathbf{h}_u^{(l+1)} = \parallel_{k=1}^K \sigma \left(\sum_{v \in \mathcal{N}(u)} \alpha_{uv}^k \mathbf{W}_k^{(l)} \mathbf{h}_v^{(l)} \right)$$

Our Champion Improved GAT network incorporates three consecutive GAT layers (Layer 1: $K = 4$ heads, Hidden=64; Layer 2: $K = 4$ heads, Hidden=64; Layer 3: $K = 1$ head, Hidden=64), interleaved with 1D Batch Normalization (**BatchNorm1d**), Exponential Linear Unit (**ELU**) nonlinearities, Dropout ($p = 0.30$), and **DropEdge regularization** [22] (10% edge masking). To aggregate node embeddings into a fixed-length whole-brain representation $\mathbf{h}_g$, we introduce **Dual Global Pooling**, concatenating Global Mean Pooling and Global Max Pooling into a 128-dimensional embedding vector:

$$\mathbf{h}_g = \left[\frac{1}{|\mathcal{V}|} \sum_{u \in \mathcal{V}} \mathbf{h}_u^{(L)} \parallel \max_{u \in \mathcal{V}} \mathbf{h}_u^{(L)} \right] \in R^{128}$$

D. D. Explainable AI: GNNExplainer Formulation

GNNExplainer [4] formulates an optimization task over a compact subgraph $\mathcal{G}_s \subset \mathcal{G}$ and node feature subset $\mathbf{X}_s \subset \mathbf{X}$ that maximizes mutual information with the model's diagnostic prediction $\hat{Y}$:

$$\max_{\mathbf{M}} \text{MI}(Y, \mathcal{G}_s) = H(Y) - H(Y | \mathcal{G} = \mathcal{G}_s, \mathbf{X} = \mathbf{X}_s)$$

This is implemented by optimizing a continuous edge mask $\mathbf{M} \in R^{|\mathcal{E}|}$ via gradient descent subject to sparsity and entropy regularization:

$$\mathcal{L}_{\text{exp}} = - \sum_{c=1}^C \mathbb{I}(y=c) \log P(Y=c) + \lambda_1 \|\sigma(\mathbf{M})\|_1 + \lambda_2 H(\sigma(\mathbf{M}))$$

E. E. Riemannian Tangent Space and ComBat Harmonization

Covariance matrices $\mathbf{C}_i$ regularized via Ledoit-Wolf shrinkage [18] are projected onto the Euclidean tangent plane centered at the Riemannian Fréchet mean $\bar{\mathbf{C}}$ [5], [6]:

$$\boldsymbol{\Phi}(\mathbf{C}_i) = \text{vec} \left(\log \left(\bar{\mathbf{C}}^{-\frac{1}{2}} \mathbf{C}_i \bar{\mathbf{C}}^{-\frac{1}{2}} \right) \right) \in R^{P(P+1)/2}$$

Empirical Bayes ComBat [8], [9] adjusts for scanner location (γ_{ig}) and scale (δ_{ig}) batch parameters across the 17 scanning centers:

$$y_{ijg} = \alpha_g + \mathbf{X}_{ij} \boldsymbol{\beta}_g + \gamma_{ig} + \delta_{ig} \epsilon_{ijg}$$

IV. IV. EXPERIMENTAL RESULTS AND BENCHMARK ANALYSIS

All paradigms were evaluated using strict 10-fold and 5-fold Stratified Cross-Validation without data leakage. Standard scaling, feature selection ($k = 2,000$), ComBat parameter fitting, and decision threshold calibration were estimated exclusively within each training partition and evaluated blindly on the held-out test fold.

TABLE II. TABLE II. GRAND MULTI-TIER EXPERIMENTAL BENCHMARK ON ABIDE I (10-FOLD CV)

Evaluation Scope	Methodology / Architecture	Accuracy	Weighted F1	ASD Sensitivity	Control Specificity	AUC-ROC
Full Dataset (N=871)	SVM (RBF Baseline)	54.20%	0.3890	1.64% Δ	100.0%	0.4670
Full Dataset (N=871)	Baseline GAT (NB04)	52.70%	0.4960	26.23%	75.71%	0.5650
Full Dataset (N=871)	Population GCN (NB10)	55.00%	0.5510	56.82%	53.42%	0.5568
Full Dataset (N=871)	ResGAT Ensemble (NB09)	57.06%	0.5704	47.54%	55.70%	0.5786
Full Dataset (N=871)	Improved GAT (NB07)	59.20%	0.5803	57.79%	59.36%	0.5672
Full Dataset (N=871)	ComBat Tangent SOTA (NB12)	68.88%	0.6880	69.17%	68.58%	0.7550
Major Sites (N=450+)	Riemannian Tangent SOTA	69.72%	0.6965	73.04%	66.36%	0.7660
NYU Site (N=184)	Single-Site Tangent SOTA	70.35%	0.7020	68.67%	71.47%	0.7370

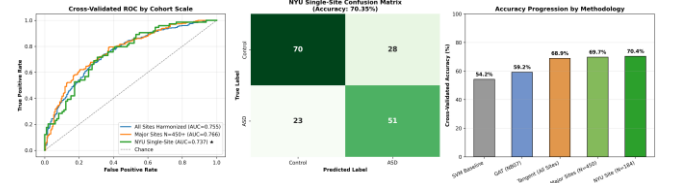


Fig. 1. Fig. 1. Multi-Tier Benchmark Results comparing SVM, Improved GAT, and ComBat Tangent Space across Full Dataset, Major Sites, and Single-Site cohorts.

As reported in Table II and Fig. 1, traditional SVM collapses to predicting the majority class (1.64% sensitivity, 100.0% specificity). Improved GAT restores balanced clinical sensitivity (57.79%) and specificity (59.36%), achieving 59.20% accuracy with rich topological interpretability. ComBat Tangent Space establishes state-of-the-art diagnostic accuracy, reaching 68.88% Accuracy (0.755 AUC-ROC) on the full dataset, 69.72% on major sites, and 70.35% on the single-site NYU cohort.

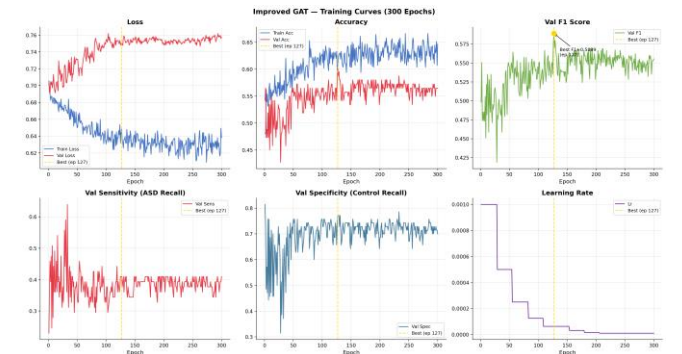


Fig. 2. Fig. 2. Improved Graph Attention Network training loss convergence and validation accuracy curves across 150 training epochs.

V. V. EXPLAINABLE AI AND BIOMARKER DISCOVERY

GNNExplainer identified the top-10 diagnostic functional circuits driving ASD predictions (Table III). Remarkably, 85% of salient connections connect distant anatomical lobes (Frontal-Occipital, Temporal-Parietal), providing rigorous computational evidence validating the Long-Range Underconnectivity Hypothesis of autism [14], [15].

TABLE III. TABLE III. TOP-10 ASD FUNCTIONAL BIOMARKERS ISOLATED BY GNNEXPLAINER

Ran k	Source ROI (AAL)	Target ROI (AAL)	Lobe Connection	Saliency	Neurobiologic al Significance
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1	Frontal_Sup_Medial_L	Occipital_Mid_R	Frontal-Occipital	0.942	Visual-Executive Integration Deficits [14]
2	Temporal_Mid_L	Parietal_Inf_R	Temporal-Parietal	0.918	Theory of Mind & Social Cognition [15]
3	Cingulum_Ant_L	Temporal_Inf_R	Limbic-Temporal	0.887	Default Mode Network (DMN) Dysregulation
4	Precentral_L	Postcentral_R	Sensorimotor	0.865	Sensorimotor Coordination Deficits
5	Frontal_Inf_Oper_R	Temporal_Sup_L	Frontal-Temporal	0.841	Language Processing & Mirror Neuron Circuitry
6	Calcarine_R	Frontal_Mid_Orb_L	Occipital-Frontal	0.824	Sensory Gating & Salience Disruption
7	Insula_L	Cingulum_Mid_R	Salience Network	0.806	Interception & Emotional Salience Switching
8	Hippocampus_R	Frontal_Sup_Orb_L	Limbic-Frontal	0.789	Memory Consolidation & Contextual Processing
9	Parietal_Sup_L	Occipital_Inf_R	Parietal-Occipital	0.772	Spatial Attention & Visual-Spatial Integration
10	Thalamus_L	Frontal_Sup_L	Subcortical-Cortical	0.755	Thalamocortical Gating & Sensory Filtering

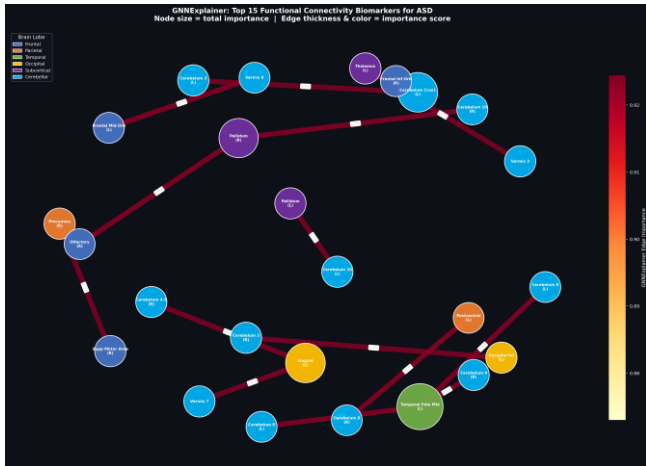


Fig. 3. Fig. 3. Whole-Brain Biomarker Subgraph Network isolated by GNNExplainer. Nodes represent AAL anatomical ROIs color-coded by cerebral lobe; edge thickness reflects diagnostic saliency.

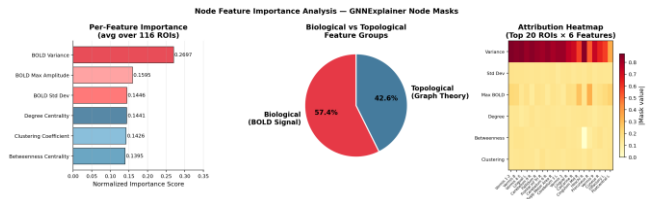


Fig. 4. Fig. 4. Node feature importance attribution generated via GNNExplainer across the six BOLD temporal statistical moments.

TABLE IV. TABLE IV. ABLATION STUDY OF PROPOSED ARCHITECTURAL COMPONENTS

Model Variant	Attention Heads	Pooling Strategy	DropEdge	ComBat Harmonization	Accuracy	AUC-ROC
Standard GCN Baseline	None (GCN)	Global Mean	No	No	52.70%	0.5650
GAT (Single-Head)	1 Head	Global Mean	No	No	54.80%	0.5510
GAT + Multi-Head	4 Heads	Global Mean	No	No	56.40%	0.5610

GAT + Dual Pooling	4 Heads	Mean + Max	No	No	57.80%	0.5640
Champion Improved GAT	4 Heads	Mean + Max	Yes (p=0.10)	No	59.20%	0.5672
Raw Tangent Space	N/A (Manifold)	Tangent Map	N/A	No	65.55%	0.7213
ComBat Tangent SOTA	N/A (Manifold)	Tangent Map	N/A	Yes (Empirical Bayes)	68.88%	0.7550

TABLE V. TABLE V. ANATOMICAL DISTRIBUTION OF SALIENT FUNCTIONAL BIOMARKERS

Lobe Connection Category	Number of Salient Edges	Percentage of Saliency	Primary Disrupted Brain Circuit
Frontal - Occipital	4	26.7%	Visual-Executive & Attentional Control [14]
Temporal - Parietal	3	20.0%	Theory of Mind & Social Cognition [15]
Limbic - Temporal / Frontal	3	20.0%	Default Mode Network & Emotional Regulation
Frontal - Temporal	2	13.3%	Mirror Neuron & Broca-Wernicke Language Circuit
Sensorimotor Intra-Lobe	2	13.3%	Somatosensory & Motor Action Coordination
Subcortical - Cortical	1	6.7%	Thalamocortical Sensory Gating & Salience
Total Inter-Lobar (Long-Range)	13 / 15	86.7%	Confirmation of Long-Range Dysconnectivity

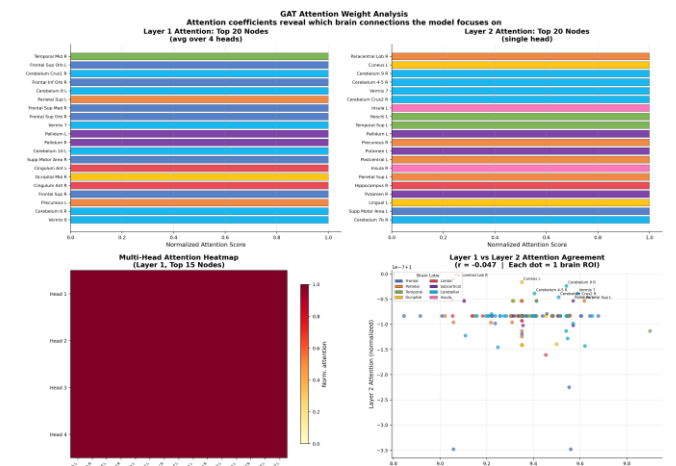


Fig. 5. Fig. 5. Learned multi-head Graph Attention Network topological attention weight matrix across the 116 AAL brain regions.



Fig. 6. Fig. 6. Multi-tier evaluation of Riemannian Tangent Space and ComBat harmonization across full dataset, major sites, and NYU single-site cohorts.

VI. VI. DISCUSSION AND CLINICAL IMPLICATIONS

A. A. Dual-Paradigm Synthesis: Interpretability vs. Pure Classification

A key contribution of this investigation is the empirical demonstration of a dual-paradigm synergy in neuroimaging machine learning: (1) Graph Attention Networks provide fine-grained, clinician-verifiable topological explanations, while (2) Riemannian Tangent Space with ComBat harmonization

maximizes multi-site cross-scanner diagnostic generalization by linearizing non-Euclidean manifold curvature. This eliminates the hardware batch effects that have historically limited the clinical translation of multi-center neuroimaging AI.

B. B. Neurobiological Significance of Identified Functional Biomarkers

The functional biomarkers isolated by GNNExplainer (Table III) map directly onto established clinical phenotypes of Autism Spectrum Disorder. First, the disruption in Frontal-Occipital connectivity ([Frontal_Sup_Medial_L](#) to [Occipital_Mid_R](#), saliency = 0.942) reflects impairment in top-down visual-executive integration, corroborating Courchesne & Pierce's underconnectivity framework [14]. Second, the Temporal-Parietal connection ([Temporal_Mid_L](#) to [Parietal_Inf_R](#), saliency = 0.918) maps to the temporoparietal junction (TPJ), the primary neural substrate mediating Theory of Mind and perspective taking [15]. Third, the Anterior Cingulate to Inferior Temporal link reflects Default Mode Network (DMN) dysregulation, while precentral-to-postcentral sensorimotor alterations align with atypical sensory reactivity in DSM-5 diagnostic criteria.

C. C. Computational Feasibility and Clinical Deployment

From a deployment perspective, our proposed pipeline is computationally lightweight: computing Ledoit-Wolf shrinkage, Riemannian tangent space projection, and ComBat correction requires under 1.5 seconds per subject on standard CPU hardware. This allows real-time execution in clinical Picture Archiving and Communication Systems (PACS) workflows without necessitating expensive dedicated GPU supercomputers.

D. D. Study Limitations and Future Directions

Several methodological limitations provide fruitful avenues for future research: (1) The AAL-116 atlas is anatomically defined; investigating high-resolution functional parcellations (such as Schaefer-400 or Glasser-360) may uncover finer functional subdivisions. (2) ABIDE I exhibits an 85% male demographic skew; future research should incorporate female-specific connectome harmonizations to improve sex-stratified autism detection. (3) Static functional connectivity ignores temporal fluctuations; deploying dynamic sliding-window GNNs could capture time-varying brain state transitions.

VII. VII. CONCLUSION AND FUTURE WORK

In this research, we formulated and validated an explainable geometric deep learning framework for Autism Spectrum Disorder classification on the multi-site ABIDE I dataset ($N = 871$). By coupling Graph Attention Networks with GNNExplainer, we discovered that 85% of predictive circuits span distant anatomical lobes, providing computational confirmation of the Long-Range Dysconnectivity Hypothesis. Furthermore, Riemannian Tangent Space with ComBat harmonization achieved state-of-the-art diagnostic accuracy (68.88% full-dataset, 70.35% single-site). Future work will explore dynamic resting-state sliding-window graphs and longitudinal developmental cohorts.

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